

System.Xml.XmlReader.Create methods

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- Article
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- 1 contributor

Feedback

This article provides supplementary remarks to the reference documentation for this API.

Most of the [Create](#) overloads include a `settings` parameter that accepts an [XmlReaderSettings](#) object. You can use this object to:

- Specify which features you want supported on the [XmlReader](#) object.
- Reuse the [XmlReaderSettings](#) object to create multiple readers. You can use the same settings to create multiple readers with the same functionality. Or, you can modify the settings on an [XmlReaderSettings](#) instance and create a new reader with a different set of features.
- Add features to an existing XML reader. The [Create](#) method can accept another [XmlReader](#) object. The underlying [XmlReader](#) object can be a user-defined reader, a [XmlTextReader](#) object, or another [XmlReader](#) instance that you want to add additional features to.
- Take full advantage of features such as better conformance checking and compliance to the [XML 1.0 \(fourth edition\)](#) recommendation that are available only on [XmlReader](#)

objects created by the static [Create](#) method.

Note

Although .NET includes concrete implementations of the [XmlReader](#) class, such as the [XmlTextReader](#), [XmlNodeReader](#), and the [XmlValidatingReader](#) classes, we recommend that you create [XmlReader](#) instances by using the [Create](#) method.

Default settings

If you use a [Create](#) overload that doesn't accept a [XmlReaderSettings](#) object, the following default reader settings are used:

Setting	Default		CheckCharacters		true		ConformanceLevel		ConformanceLevel.Document
			IgnoreComments		false		IgnoreProcessingInstructions		false
			IgnoreWhitespace		false		LineNumberOffset		0
			LinePositionOffset		0		NameTable		null
			DtdProcessing		Prohibit		Schemas		An empty XmlSchemaSet object
			ValidationFlags		ProcessIdentityConstraints enabled		ValidationType		None
			XmlResolver		null				

Settings for common scenarios

Here are the [XmlReaderSettings](#) properties you should set for some of the typical XML reader scenarios.

Requirement	Set		Data must be a well-formed XML document.		ConformanceLevel to Document .
			Data must be a well-formed XML parsed entity.		ConformanceLevel to Fragment .
			Data must be validated against a DTD.		DtdProcessing to Parse ValidationType to DTD .
			Data must be validated against an XML schema.		ValidationType to Schema Schemas to the XmlSchemaSet to use for validation. Note that XmlReader doesn't support XML-Data Reduced (XDR) schema validation.
			Data must be validated against an inline XML schema.		ValidationType to Schema ValidationFlags to ProcessInlineSchema .
			Type support.		ValidationType to Schema Schemas to the XmlSchemaSet to use.

[XmlReader](#) doesn't support XML-Data Reduced (XDR) schema validation.

Asynchronous programming

In synchronous mode, the [Create](#) method reads the first chunk of data from the buffer of the file, stream, or text reader. This may throw an exception if an I/O operation fails. In asynchronous mode, the first I/O operation occurs with a read operation, so exceptions that arise will be thrown when the read operation occurs.

Security considerations

By default, the [XmlReader](#) uses an [XmlUrlResolver](#) object with no user credentials to open resources. This means that, by default, the XML reader can access any location that

doesn't require credentials. Use the [XmlResolver](#) property to control access to resources:

- Set [XmlResolver](#) to an [XmlSecureResolver](#) object to restrict the resources that the XML reader can access, or...
- Set [XmlResolver](#) to `null` to prevent the XML reader from opening any external resources.

Examples

This example creates an XML reader that strips insignificant white space, strips comments, and performs fragment-level conformance checking.

```
XmlReaderSettings settings = new XmlReaderSettings();
settings.ConformanceLevel = ConformanceLevel.Fragment;
settings.IgnoreWhitespace = true;
settings.IgnoreComments = true;
XmlReader reader = XmlReader.Create("books.xml", settings);
```

```
Dim settings As New XmlReaderSettings()
settings.ConformanceLevel = ConformanceLevel.Fragment
settings.IgnoreWhitespace = true
settings.IgnoreComments = true
Dim reader As XmlReader = XmlReader.Create("books.xml", settings)
```

The following example uses an [XmlUrlResolver](#) with default credentials to access a file.

```
// Set the reader settings.
XmlReaderSettings settings = new XmlReaderSettings();
settings.IgnoreComments = true;
settings.IgnoreProcessingInstructions = true;
settings.IgnoreWhitespace = true;
```

```
' Set the reader settings.
Dim settings as XmlReaderSettings = new XmlReaderSettings()
settings.IgnoreComments = true
settings.IgnoreProcessingInstructions = true
settings.IgnoreWhitespace = true
```

```

// Create a resolver with default credentials.
XmlUrlResolver resolver = new XmlUrlResolver();
resolver.Credentials = System.Net.CredentialCache.DefaultCredentials;

// Set the reader settings object to use the resolver.
settings.XmlResolver = resolver;

// Create the XmlReader object.
XmlReader reader = XmlReader.Create("http://ServerName/data/books.xml", settings);

```

```

' Create a resolver with default credentials.
Dim resolver as XmlUrlResolver = new XmlUrlResolver()
resolver.Credentials = System.Net.CredentialCache.DefaultCredentials

' Set the reader settings object to use the resolver.
settings.XmlResolver = resolver

' Create the XmlReader object.
Dim reader as XmlReader = XmlReader.Create("http://ServerName/data/books.xml", settings)

```

The following code wraps a reader instance within another reader.

```

XmlTextReader txtReader = new XmlTextReader("bookOrder.xml");
XmlReaderSettings settings = new XmlReaderSettings();
settings.Schemas.Add("urn:po-schema", "PO.xsd");
settings.ValidationType = ValidationType.Schema;
XmlReader reader = XmlReader.Create(txtReader, settings);

```

```

Dim txtReader As XmlTextReader = New XmlTextReader("bookOrder.xml")
Dim settings As New XmlReaderSettings()
settings.Schemas.Add("urn:po-schema", "PO.xsd")
settings.ValidationType = ValidationType.Schema
Dim reader As XmlReader = XmlReader.Create(txtReader, settings)

```

This example chains readers to add DTD and XML schema validation.

```
XmlReaderSettings settings = new XmlReaderSettings();
settings.ValidationType = ValidationType.DTD;
XmlReader inner = XmlReader.Create("book.xml", settings); // DTD Validation
settings.Schemas.Add("urn:book-schema", "book.xsd");
settings.ValidationType = ValidationType.Schema;
XmlReader outer = XmlReader.Create(inner, settings); // XML Schema Validation
```

```
Dim settings As New XmlReaderSettings()
settings.ValidationType = ValidationType.DTD
Dim inner As XmlReader = XmlReader.Create("book.xml", settings) ' DTD Validation
settings.Schemas.Add("urn:book-schema", "book.xsd")
settings.ValidationType = ValidationType.Schema
Dim outer As XmlReader = XmlReader.Create(inner, settings) ' XML Schema Validation
```

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